Python, SQL va Django olimpiadasi

Django asoslari

Model · URL · View · Template · Form

MVT arxitekturasi

Qism	Vazifasi	Fayl
Model	Ma’lumotlar bazasidagi jadval	models.py
View	So‘rovni qayta ishlaydi	views.py
Template	HTML sahifa	templates/
URL	Manzilni view bilan bog‘laydi	urls.py
Sozlamalar	DATABASES, INSTALLED_APPS	settings.py

Model va migratsiya

```
# models.py
class Post(models.Model):
  title = models.CharField(max_length=100)
  body = models.TextField()

# admin.py
admin.site.register(Post)
```

Model
Jadvalni ifodalovchi asosiy sinf

makemigrations
Migratsiya faylini yaratadi

migrate
Migratsiyani bazaga qo‘llaydi

URL va View

```
# urls.py
urlpatterns = [
  path('posts/', views.post_list),
]

# views.py
def create(request):
  return redirect('post_list')
```

urls.py
URL manzillar shu faylda belgilanadi

View
So‘rovni qabul qilib, javob qaytaradi

redirect()
Boshqa sahifaga yo‘naltiradi, render() esa shablonni ko‘rsatadi

QuerySet va ORM

```
posts = Post.objects.filter(
  title__contains='SQL'
).order_by('-id')

for p in posts:
  print(p.title)
```

Lazy so‘rov

Baza faqat natija kerak bo‘lganda (masalan, for siklida) so‘raladi

filter()

SQL’dagi WHERE ga mos keladi

order_by('-id')

SQL’dagi ORDER BY ... DESC ga mos keladi

Template va Form

```
{% extends 'base.html' %}
{% block content %}
  <form method='post'>
    {% csrf_token %}
    {{ form.as_p }}
    <button>Saqlash</button>
  </form>
{% endblock %}
```

extends
Boshqa shablondan meros oladi

csrf_token
Formani CSRF hujumidan himoya qiladi

form.is_valid()
View’da yuborilgan ma’lumotlarni tekshiradi

Olimpiada testidan savollar

№	Savol	To‘g‘ri javob
3	Bazadagi jadvalni ifodalovchi sinf	Model
6	Tayyor migratsiyani bazaga qo‘llash	migrate
12	Shablondan meros olish	{% extends %}
15	Form ma’lumotini tekshirish	is_valid()
30	Formaga CSRF himoyasi qo‘shish	{% csrf_token %}